



Learning with large language models: beyond prompt engineering

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Abstract

With the advancement and increased accessibility of artificial intelligence, particularly large language models (LLMs), LLMs have garnered attention in education and are postulated to hold great promise in supporting learning. Nevertheless, these promises come with many challenges. The foremost challenge is the lack of awareness of LLMs' potential and the misuse of these tools. Researchers and educators have proposed frameworks and conducted studies to address these challenges, such as teaching prompt engineering (PE) to students to enhance their interactions with LLMs. However, we take a step further and propose the HIIC model (**H**uman to **I**nitiate, **I**nteract, and **C**ritique), arguing that learning with LLMs requires not only PE techniques but also the ability to ask additional questions and make warranted judgments about LLM-generated responses. In this study, students enrolled in a physics class were instructed to converse with ChatGPT (an LLM) about their incorrect answers on an exam. We utilized the ICAP framework (Interactive, Constructive, Active, and Passive) to analyze the additional questions students asked ChatGPT and examined the strategies they used to evaluate ChatGPT's responses. The results showed that students had limited engagement with ChatGPT, meaning they either did not ask any additional questions or asked only surface-level ones, such as fact-checking. Moreover, students relied heavily on anecdotal experiences or provided no reference when evaluating ChatGPT's responses. However, students who were more cognitively engaged with prompting showed higher learning gains than those who only superficially engaged in prompting questions. These findings highlight the need for student training that goes beyond teaching PE to focus on interactive engagement with LLMs, rather than using LLMs merely to complete tasks (which may contribute to plagiarism concerns). The implications and limitations of the study are also discussed at the end.

Keywords ICAP · Large Language Models · Learning · Prompt Engineering · Critical Thinking

1 Introduction

The educational value of emerging technologies depends not only on their technical capabilities but also on how they shape learners' cognitive engagement, metacognitive monitoring, and self-regulated inquiry. Decades of research have demonstrated that meaningful learning is fostered when students actively generate questions, monitor their understanding, and iteratively refine their ideas through inquiry-based interactions, rather than passively receiving information. As new artificial intelligence (AI) tools enter educational contexts, an important challenge is understanding how learners actually engage with these tools in relation to established principles of active cognitive engagement and self-regulated learning.

With the debut of OpenAI ChatGPT in November 2022, generative AI (GenAI) has overhauled the landscape of educational practices (Lo, 2023). As a prominent example of GenAI, ChatGPT highlights the potential of large language models (LLMs) to enable personalized, interactive learning by allowing students to explore concepts, ask questions, and engage in self-directed inquiry. Due to the ability of LLMs to perform a wide range of tasks, educators have suggested that they can be used as a tool to support teaching in a wide range of subjects, including business (Alshater, 2022), engineering (Qadir, 2023), and nursing education (O'Connor, 2022). LLMs have been proposed to have the potential to support adaptive learning, automated essay grading, personalized tutoring, etc. (Baidoo-Anu & Ansah, 2023). Theoretical frameworks for how to use LLMs for teaching and learning have been proposed, and studies have been conducted to teach prompt engineering (PE) techniques, the practice of crafting inputs to guide a GenAI system toward more useful or accurate outputs, to students (Cain, 2024; Su & Yang, 2023).

While there is a mix of both excitement and skepticism toward using LLMs in supporting student learning, many students have already been using LLMs for their learning; however, in both correct and incorrect ways. Research has shown that students held a limited understanding of LLMs, often resulting in accepting the information at face value without making informed judgments or knowing how to ask meaningful questions to LLMs to assist in their learning (Ding et al., 2023). Teaching students to use LLMs effectively for their learning is a multifaceted task that goes beyond simply teaching PE techniques. The ultimate goal of PE is to seek the desired answers from GenAI, while learning often requires students to collaborate with GenAI, and the ultimate goal is usually beyond simply seeking answers, but having a comprehensive understanding of a certain topic. Furthermore, due to the infamous hallucination effect of LLMs, it also involves training students to have a critical mindset to make a warranted judgment of the information or products generated by GenAI (Ferrara, 2024). To develop practical guidelines that foster responsible and meaningful interactions with LLMs, it is critical to first understand how students engage with these tools when it comes to supporting their learning.

In this study, we frame student interactions with LLMs through the lens of ICAP theory (Chi, 2009; Chi & Wylie, 2014), which posits that learning outcomes improve as learners move from passive to interactive engagement. ICAP theory has been widely used to examine learners' cognitive engagement in digital learning environments, including video-based instruction, technology-supported inquiry, and text pro-

duction or explanation activities, where differences in observable learner behaviors reflect underlying levels of cognitive processing. These prior applications provide a theoretical basis for using ICAP to analyze students' interaction patterns with LLMs. Specifically, the act of generating questions while using an LLM can be seen as either an active or constructive learning behavior, depending on how learners engage with the content.

Despite the growing availability of LLMs in educational settings, little is known about how students generate and refine their questions when interacting with such tools, or how these questioning strategies reflect deeper cognitive engagement (Elim, 2024). Existing literature has largely emphasized ethical considerations, policy and management implications, and broad applications of AI in education, with comparatively less attention to students' moment-to-moment inquiry processes, cognitive engagement, and interaction patterns during learning with LLMs (Yu et al., 2025). By applying the ICAP framework, we examined how students' questioning behaviors aligned with different levels of cognitive effort, providing a theoretical rationale for analyzing how learners interact with AI to support their own understanding. The study was conducted with 40 students enrolled in a physics class, and the data were collected from students' questions asked to ChatGPT to revisit the incorrectly answered questions on an exam. To this end, we aim to answer these three research questions (RQs):

RQ1. What types of questions do students ask LLMs to facilitate interactions?

RQ2. How do different types of questions students ask LLMs influence their learning outcomes?

RQ3. What strategies do students use to evaluate the information received from LLMs?

2 Background

2.1 Prompt engineering

Humans rely on prompts to communicate with LLMs. PE is a relatively recent discipline that focuses on developing and optimizing prompts to effectively utilize LLMs to produce desired, relevant, and accurate outputs (Brown et al., 2020). This involves crafting the prompt in a way that maximizes the model's performance, aligns with the task, and elicits the most useful responses. The term "Prompt Engineering" originally came from a popular post online about GPT3 (an LLM) and was formalized by Liu et al. (2023).

PE plays a crucial role in optimizing the interaction with LLMs. Essentially, clear and specific prompts are more likely to elicit more accurate responses per the inquiries intended, and more descriptive verbs are more likely to stimulate more divergent results (Bowen & Watson, 2025). PE techniques take on many forms, such as zero-

shot, one-shot/few-shot, self-consistency, tree-of-thoughts, and multi-turn (Chen et al., 2023; Ein-Dor et al., 2024). The primary goal of these techniques is to maximize the performance of LLMs to generate intended outcomes. For instance, in zero-shot prompting, a user might ask the model to perform a task it has not explicitly been trained for. In contrast, one-shot or few-shot prompting provides the model with a few examples to help it generate more relevant responses. As highlighted by Wei et al. (2022), chain-of-thought prompting encourages the LLMs to reason step-by-step through a problem, thus increasing the accuracy of complex answers. Multi-turn prompting, on the other hand, allows the LLM to engage in multi-round prompt refinement, refining its responses based on feedback from the user. These techniques ensure that the model produces high-quality and contextually appropriate outputs. At the same time, the iterative refinement in the multi-turn technique is critical in addressing issues such as hallucinations (Johnson, 2024).

PE is increasingly being called attention to in educational settings to enhance learning experiences, which is believed to hold great promise for educators and students alike. Eager and Brunton (2023) emphasize that educators must become proficient in designing prompts that align with pedagogical objectives, enabling the creation of customized content and assessments. For example, using few-shot learning, educators could provide an LLM with examples of quiz questions and instruct it to generate similar content for their class. In addition, educators can apply chain-of-thought prompting to generate feedback that guides students through complex problems step by step, improving their understanding. Bozkurt and Sharma (2023) further discuss how educators can leverage PE to generate resources, such as lecture notes, case studies, and even role-playing scenarios for classroom exercises. In addition, PE has also been proposed to be used to foster deeper critical thinking by prompting students to reflect on the biases or limitations inherent in AI-generated content (Davis, 2025).

2.2 LLMs and student learning

Research exploring the impact of LLMs on student learning has yielded mixed results in terms of their variability in students' engagement, academic performance, and motivation (Chiu et al., 2023; Grassini, 2023; Selwyn, 2022; Seo et al., 2021; Somasundaram et al., 2020). For instance, a study led by Guo et al. (2025) in STEM education found that LLM integration significantly improved middle school students' scientific knowledge, attitudes, and problem-solving skills. Whereas another study by Jost et al. (2024) in programming education revealed that an increase in students' reliance on LLMs for critical thinking-intensive tasks, such as code generation and debugging, correlated with lower grades amongst students. Similarly, Lehmann et al. (2024) in two pre-registered and incentivized laboratory experiments found that LLMs had no effect on learning outcomes per se.

The exploratory analyses, however, revealed that students' prior knowledge and LLM usage behavior affected their learning with LLMs. For instance, students who asked LLMs for explanations of topics had been found to have improvement in their understanding of the topic, whereas students who asked LLMs for "solutions" to practice problems were able to cover a breadth of topics, but the depth of their understanding was seemingly less. Furthermore, the study revealed that students with less

prior subject knowledge learned less with LLMs in comparison to students with a high level of prior knowledge. This only highlights the inherent complexity of using LLMs in educational settings, as they have the potential to take on two distinct roles in the learning process: complementation and substitution. Although both are found to support learning, students who use LLMs to complement their learning (asking for explanations) aid in deepening their understanding, whereas students who use LLMs as substitutes (asking for solutions) aid in covering more topic areas or volume of study material (Lehmann et al., 2024). Hence, LLMs such as ChatGPT might be beneficial to the learning process, but they also run the risk of supporting learning at a superficial level and stifling critical thinking, problem-solving, and creativity skills (O'Connor, 2023). Both studies emphasize that LLMs in student learning are more beneficial when used as a supplementary learning tool as opposed to a complementary aid (Jost et al., 2024; Lehmann et al., 2024).

In an effort to integrate LLMs into higher education, Zdravkova et al. (2024) compare three approaches: traditional or manual assignment preparation without using LLMs support, complete reliance on LLMs, and a hybrid approach combining both. The study conducted with undergraduate students examined preferences, satisfaction, productivity, and comfort with these approaches across two assignment types—essay writing and programming tasks. The findings revealed that students favored the hybrid approach over both traditional preparation and full reliance on LLMs. The majority of the students viewed LLMs as a valuable tool for completing assignments, as long as they engaged critically with the LLM-generated output, stressing the importance of human oversight and engagement in the process. Similarly, Lepp and Kaimre (2025) investigated students' use of AI chatbots in a programming course, examining the relationship between LLM usage and academic performance. Based on a student survey that analyzed the frequency and manner of LLMs use, their findings indicated a negative correlation between students' academic performance and the reported frequency of chatbot tools usage, suggesting that students who used AI tools more frequently tended to struggle more with coursework. One possible explanation is that low-performing students, who experience difficulty with course material, may rely more on AI as a supplementary aid, an observation that aligns with previous research indicating that low-performing students tend to use additional resources more frequently (Lepp & Kaimre, 2022). The study stresses the need to mark a distinction in students' LLM usage patterns, suggesting it remains supportive rather than central to students' learning process. Furthermore, these findings underscore the need for structured guidelines and a pedagogical framework to support effective integration of LLMs and AI tools in education, one that fosters students' deeper learning and prevents over-reliance and dependency on LLM technologies (Lepp & Kaimre, 2025).

2.3 ICAP framework as an analytic lens for human-LLM dialogue

The ICAP framework was originally proposed by Chi (2009) to analyze data to differentiate student engagement in learning, and was then widely used in research for student observable learning behaviors, with instructions along with student-produced outputs (e.g., notes or questions) (Chi & Boucher, 2023). The ICAP framework cat-

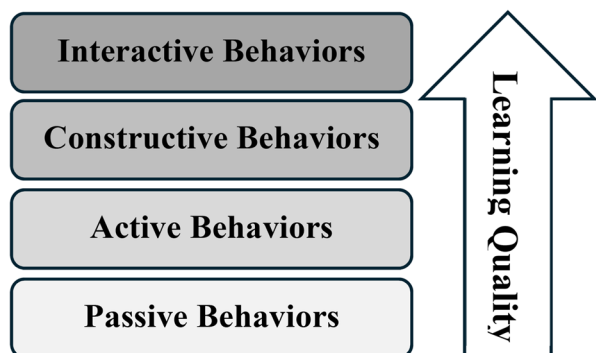
egorizes student cognitive engagement in learning into four modes: Interactive, Constructive, Active, and Passive, and it emphasizes “minds-on” rather than “hands-on” (Chi et al., 2018).

The **interactive** mode refers to a learning process where two or more students exchange understanding, infer and create meaning from knowledge, and generate new ideas together. In the **constructive** mode, students infer and generate new ideas beyond the provided learning materials individually. The **active** mode represents a learning process in which students demonstrate a predefined set of learning behaviors without generating new ideas. When students are in **passive** mode, they receive information without exhibiting learning behaviors. Student engagement in interactive and constructive modes highlights the **generativeness** of ideas, while students engaged in active and passive modes simply follow the instructions.

For instance, a student watching a lecture video can be hands-on engaged because they follow the instructions to watch videos, but they are not necessarily minds-on if they are not taking notes or participating in discussions about the content. In fact, if the student only watches the video without demonstrating any learning behaviors, it is classified as passive learning according to ICAP; however, if the student generates notes while watching the video or participating in an online discussion about the content of the video, it can be classified as constructive or interactive, respectively. The ICAP framework postulates that these modes follow a hierarchical order: Interactive > Constructive > Active > Passive (see Fig. 1). The quality of student learning is dependent on the mode students engage in during learning activities (Menekse et al., 2013). Students who receive didactic-lecture-style instruction are more likely to exhibit passive or active behaviors, resulting in a lower level of learning (Freeman et al., 2014).

The ICAP framework has long been shown to be effective in guiding educators in designing instructional strategies that promote deeper cognitive engagement and improve learning outcomes. For example, rather than asking students to memorize facts or definitions (Passive or Active engagement), educators can ask questions that require students to analyze, infer, and generate new ideas, such as “How does this concept apply in a real-world scenario?” or “What would happen if we altered this variable?” (Ding et al., 2024; Morris & Chi, 2020). These types of questions push students into constructive engagement, where they must actively generate new con-

Fig. 1 ICAP Framework. The quality of a student’s learning increases with each level of the student’s cognitive engagement



nections or extend existing knowledge, resulting in better student performance than those questions that require no generative information from students. Another example in which teachers were trained to incorporate more interactive activities, such as group discussions and peer feedback, that encourage students to build on each other's ideas and engage in back-and-forth reasoning, has been shown to generate better learning performance than among the students who participated in the other three modes of learning (Chi, 2015). According to ICAP, questions that contain generative information are more likely to improve learning than those without. In this study, we used ICAP to analyze students' questions asked of an LLM.

3 Proposed conceptual framework on prompt for learning

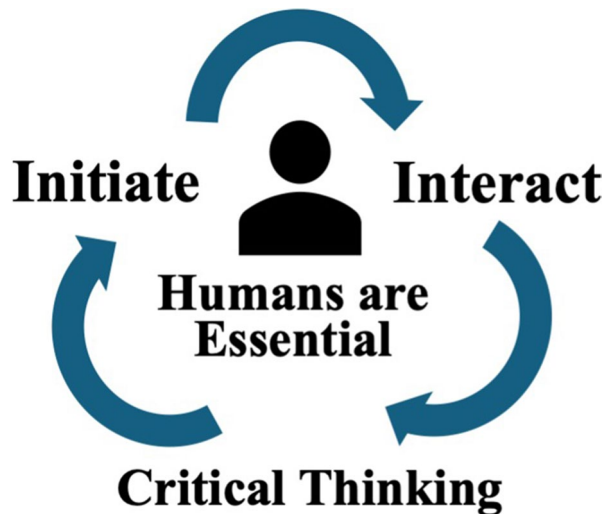
Several key differences distinguish teaching students PE techniques from teaching them to learn with LLMs. First, regardless of how many or what PE techniques a student employs to interact with LLMs, the aim is to prompt LLMs to generate the answers that are closest to what the student requires the LLMs to perform. Learning, however, is not merely about receiving correct information; it is a process of actively building knowledge. Engaging with LLMs allows students to explore information related to a concept and integrate it into their existing understanding through iterative interactions. Second, because learning is an ongoing process, engaging with LLMs requires continuous interaction to expand knowledge (Chi & Wylie, 2014). In contrast, PE emphasizes crafting effective prompts to generate specific outcomes, which can often be achieved within a single exchange. Finally, LLMs can produce responses that appear plausible but contain errors (i.e., hallucination effect), depending on factors such as the quality of the training data or the precision of the prompts (Farquhar et al., 2024). While such errors may indicate a failed attempt in PE, in learning contexts, they present valuable opportunities for reflection and growth. Identifying, analyzing, and correcting these faulty responses can help students deepen their understanding and foster critical thinking (Ding et al., 2021).

Learning with LLMs differs from discussions in conventional learning environments, where educators typically initiate and guide conversations, whether teacher-led or student-centered, and online or in-person (Hennessy et al., 2016). In these settings, educators act as both facilitators and authorities on the accuracy of the information being discussed. In contrast, learning with LLMs shifts the ownership of learning to students, who need to take on the roles of both initiators and gatekeepers of the information they receive. Simply teaching PE techniques is not enough and can even pose risks in this context. LLMs can provide plausible but inaccurate information, and thus, the most essential skill students need to develop is critical thinking. They should be able to critically analyze information from LLMs and seek reliable sources to verify its validity (Ding et al., 2023). Table 1 summarizes the differences between prompting for learning, prompt engineering, and discussions in traditional educational settings.

Given the vast differences between prompting for learning, PE, and traditional education, students need a distinct set of skills to effectively engage with LLMs for learning. To begin addressing this need, we propose the HIIC model (Fig. 2) as a

Table 1 Differences between prompting for learning, PE, and discussions in traditional education

Goal	Prompt Engineering	Prompt for Learning	Traditional Discussions
	Accurate Output	Knowledge Building	Knowledge Building
Initiator	Students	Students	Educators
Facilitator	Students	Students	Educators
Authority	LLMs	Students	Educators
Process	One Round	Interactive Rounds	Interactive Rounds
Errors	Consider as Failure	Learning Opportunity	Learning Opportunity

Fig. 2 The conceptual framework of the proposed HIIC model

preliminary conceptual framework to guide our analysis. It emphasizes that humans are essential in the interaction with LLMs and identifies three essential competencies: Initiating, Interacting, and Critiquing. First, a student needs to initiate the conversation with an LLM. This step requires the student to be familiar with basic PE techniques to start the conversation. Then, the student will need to know how to ask follow-up questions to interact with the LLM. In this step, students need to play the role of a facilitator in the dialogue, which requires them to be equipped with the techniques of how to ask high-quality questions to LLMs to expand their knowledge. The follow-up questions that the students ask can go beyond the original question and are based on the outputs from the LLMs. Therefore, this phase is an interactive collaborative dialogue between the students and LLMs. Knowing how to ask questions is not sufficient when interacting with LLMs due to the “hallucination” effect of LLMs; one needs to have a critical thinking mind to be able to make a warranted judgment of the information one receives from the LLMs. Finally, whenever it reaches a point of whether to accept the information provided by the LLMs, it is always humans who should decide whether to accept the response or challenge and further the conversations (Yang et al., 2024).

4 Methodology

4.1 Research context

This study involved 40 undergraduate students enrolled in an introductory college-level physics course at a public university in the southern United States. Most participants identified as female, reported no prior experience with ChatGPT, and participated voluntarily. Participants received 10 bonus points for their participation. Among the participants, 36 (90%) self-identified as female, 3 (7.5%) as male, and 1 (2.5%) as non-binary. The majority of the participants were Caucasian/White ($n=29$, 72.5%), followed by African American/Black ($n=9$, 22.5%), with 1 participant each identified as Asian and Native American. Participants' ages ranged from 19 to 38, with an average age of 21.26 years. Almost all participants reported having no prior experience with ChatGPT, except for 3 individuals who had used it briefly for testing or occasional purposes.

The course was conducted over a 16-week semester and covered three key physics concepts, each assessed through an exam. This study focused on the second exam, which consisted of 50 multiple-choice questions designed to evaluate participants' understanding of topics such as light, radioactivity, and related concepts. We selected the second exam for this study because it provided the most appropriate timing for implementing the study. The first exam occurred early in the semester, when students were still adjusting to the course and workload, and the third exam was scheduled close to the final exam period, when students were likely to be too busy or stressed to fully engage with our study. Thus, Exam 2 offered the most stable and feasible opportunity to observe students' interactions with ChatGPT in a low-stakes learning context. Participants completed the exam during an in-person class session, with responses graded as either correct or incorrect. Following the exam, participants were given a week and a half to complete a makeup assignment, allowing them to regain lost points by "chatting" with ChatGPT to review and address their incorrect responses. This study was approved by the university's Institutional Review Board (IRB). Informed consent was obtained from all participants prior to data collection. To protect confidentiality, all data were anonymized, and identifying information was removed before analysis.

4.2 Research design

The makeup exam assignment was structured as a fillable form, with participants allowed to answer up to 10 questions from their original exam. The first page of the assignment outlined a four-step set of instructions, which included guidance on creating a ChatGPT account and detailed steps to complete the assignment for each question they incorrectly answered on the exam (Fig. 3). Participants were also informed that ChatGPT is an Artificial Intelligence system prone to errors and explicitly encouraged to critically evaluate the responses generated by ChatGPT. In addition, the exam key was made available to participants prior to the assignment of the makeup exam.

Step 1: Go to: <https://chat.openai.com/auth/login>

Step 2: Log in if you already have an account; sign up if you do not have an account. (Here is a brief tutorial on how to access ChatGPT and what it is: <https://youtu.be/t1-5z0HgkuE>. Directly go to 1:20 of this video to learn how to use it.)

Step 3: Locate the questions that you answered incorrectly in your Exam 2.

Step 4: Complete each question by filling out this form. This form allows you to complete a maximum of 10 questions, if you do not wish to make up points for all of them, you can simply leave them blank.

Use all the prompts in this form for each of the questions you want to address. However, you are not limited to the provided prompts, you can ask ChatGPT as many questions as you want to address the questions.

NOTE: ChatGPT is an artificial intelligence system, it can make mistakes and can give you wrong answers. Think critically when you receive an answer from ChatGPT.

Fig. 3 The instructions for the makeup assignment

For each question, participants were instructed to follow a specific sequence when interacting with ChatGPT:

1. Participants posed the original exam question to ChatGPT and received its response.
2. Participants compared their original incorrect answers with ChatGPT's response:
 - If their answers matched ChatGPT's, they would decide whether they agreed or disagreed with ChatGPT's answer and provide a rationale for their decision.
 - If their answers did not match one of ChatGPT's, they would share their original incorrect answers with ChatGPT, evaluate ChatGPT's following response, and decide whether they agreed or disagreed with a provision of the rationale for their decision.
3. Participants were encouraged to ask additional questions of ChatGPT during the entire interaction.

4.3 Analyses

To address the first RQ regarding the types of questions students asked to facilitate interactions with LLMs, the additional questions (the third step in the sequence) posed by students to ChatGPT were coded. We chose only to analyze the additional questions because the original questions students asked ChatGPT would only follow the instructions without any agentic behaviors; however, the goal of this study was to investigate students' initiation of interactions with LLMs. To ensure rigor,

we followed an iterative process and used both deductive and inductive methods to develop the coding book (Bingham, 2023). Deductively, guided by the ICAP framework, the first author developed an original coding rubric for the analysis of student questions. Inductively, two steps were taken. First, two graduate research assistants (GRA) were randomly assigned 20% of the student questions to apply the rubric for the analysis. They applied the rubric to the questions independently and revised the descriptions of each mode as needed to adapt to the data. The two GRAs then met with the first author to discuss discrepancies in their independent rubrics and analyses of the questions until an agreement was reached. The main discrepancies were for the constructive questions. Often, these questions were confused with active questions because constructive questions should contain ideas generated by students rather than direct information-seeking. For example, “*What is it called when waves combine and produce a standing wave?*” could be easily confused with a constructive question, but it was actually an active question because the “*waves combine and produce a standing wave*” in the question was from a distractor of the exam question. The meetings between the two GRAs and the first author primarily aimed to address these disagreements and establish strategies for easily categorizing the participants’ questions. Using the updated rubric, the two GRAs independently coded the same sample data again. A second meeting was held with the first author to address any remaining differences and observations, ensuring that the descriptions of each mode fully captured the nuances of the data and were aligned with the ICAP framework. Then, one of the GRAs coded the rest of the data using the final version of the rubric.

To investigate the second RQ, which is how interacting with LLMs influenced students’ learning, we first conducted a clustering analysis by using students’ frequencies of Passive, Active, Constructive, and Interactive questions derived from RQ1, and then performed a repeated measures ANOVA on students’ learning changes from Exam 2 to the final exam. The purpose of these two-step analyses was to investigate how the different engagement modes may have resulted in different students’ learning. The final exam contained 80 multiple-choice questions covering the concepts from all three exams. To ensure alignment, the course instructor identified 26 questions from the final exam that specifically assessed the same concepts covered in Exam 2. We then calculated the percentage of correct answers from those 26 questions and scaled that percentage to a score out of 50, making it directly comparable to the Exam 2 score.

To address the third RQ about the strategies that students used to evaluate the information provided by the LLM, an intuitive analysis was applied. That is, the two GRAs independently analyzed the data 20% and generated a coding book of the critical thinking strategies that the students used to rationalize their judgments on whether or not to agree or disagree with the answers provided by ChatGPT. The two then met with the first author to discuss discrepancies in their coding book until an agreement was obtained. One GRA used the coding book to analyze the rest of the data; however, when new strategies appeared in the data that were not captured in the coding book, the strategies were added.

5 Results

5.1 RQ1. What types of questions do students ask LLMs to facilitate interactions?

Based on the ICAP framework, students' additional questions were classified into four modes. Table 2 presents both the final coding rules and representative examples for each category. According to ICAP, the passive mode (passively receiving information) and active mode (following instructions to act) behaviors do not require students to generate additional ideas. In contrast, the constructive and interactive modes typically involve students producing generated information after receiving instruction. Therefore, in our rubric, the **Passive** mode was one in which students did not pose any additional questions to ChatGPT after they received the feedback from ChatGPT to the incorrectly answered questions on their exam. The questions that fell into the **Active** mode were those questions that simply sought a definition of a term

Table 2 Developed rubric based on ICAP for student questions to ChatGPT

Mode	Keywords	Descriptions	Examples
Passive	No additional questions	No additional questions were asked after receiving responses.	<i>N/A</i>
Active	No generated insights	Questions asking for straightforward facts or definitions of terms in the stem or distractors of an exam question. Inquiry why the distractors are not correct.	<i>What is the definition of a heat engine? What is it called when waves combine and produce a standing wave? (Note: this is a distractor.)</i>
Constructive	Contain generated insights	Questions that require students to make connections between concepts, express curiosity about a concept, or recall prior knowledge that goes beyond the information in the exam questions.	<i>What if I'm touching a capacitor instead of a battery? Does Celsius start from an absolute temperature scale?</i>
Interactive	Build upon responses with generated insights	Questions that build upon or expand upon a previous answer and seek further explanations, leading to a conversation.	<i>Q1: How can water boil without a heat source? Q2: What are additional ways of heating water? Q3: How to heat liquids without a stove?</i>

Table 3 ICAP Question Type Descriptive Statistics by Group

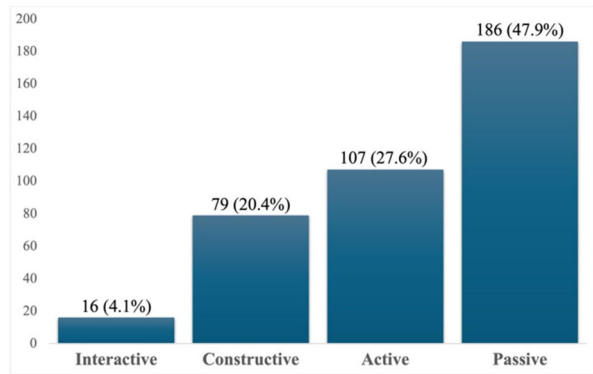
ICAP Mode	Group 1 Mean (SD)	Group 2 Mean (SD)
N of students	<i>N</i> =23	<i>N</i> =17
Passive	7.26 (3.03)	1.00 (1.28)
Active	0.70 (0.93)	5.06 (2.33)
Constructive	0.91 (1.56)	3.41 (2.85)
Interactive	0.52 (1.16)	0.24 (0.44)

appearing in the stems or distractors of the exam questions, or inquired why the other distractors were not correct. These questions mostly did not require students to add any insights or require much “mind movement,” but rather a straightforward fact check for terminology or why some distractors were not correct. The **Constructive** questions refer to those that require students to go beyond the exam questions and add their own understanding or curiosity to their questions. For instance, as shown in Table 2 for the constructive mode, the student asked, “What if I’m touching a capacitor instead of a battery?” in response to the original exam question, “When touching across a high voltage drop with two hands, you have a greater risk of electric ‘shock’ if ____.” The student’s additional question extended beyond the scope of the exam question itself, seeking other ways that could cause an electrical shock. The **Interactive** mode questions referred to those that contained generated information as the questions in the constructive mode; however, these questions also led to follow-up questions, demonstrating a back-and-forth interaction with ChatGPT.

To analyze the students’ additional questions, we first needed to decide how to chunk the questions as units for analysis. In other words, interactive mode questions usually contain several questions in a unit to complete the conversation with ChatGPT. As suggested by Chi et al. (2018, 2017), in which ICAP was used to analyze peer dialogues, a unit for analysis needs to be around the same concept, and it should be considered as a new unit when the conversation steers to a new concept. For instance, two additional questions asked by a student for the same exam question: (1) Why do isotopes have a different number of neutrons? (2) Why don’t isotopes have a different number of protons? These two questions were coded as two active questions since they did not inherently form a flow of conversation. Additionally, there were occasions when students simply expressed appreciation, such as “Thank you!” to ChatGPT’s answers, and we found that those responses were more relevant to emotional expressions rather than cognitive engagement. Therefore, we excluded them from the analysis, which were two.

Responses for a total of 383 exam questions from students were analyzed. Of these, 186 exam questions did not have additional questions posed by students, and they were coded as Passive mode (48.6%). This left 197 exam questions with additional questions, and there were 220 questions (as 202 analysis units) asked by students. Among the 202 analysis units, 107 were in Active mode (27.6%), 79 were in Constructive mode (20.4%), and 16 were in Interactive mode (4.1%). Figure 4 illustrates the frequencies of the additional questions asked by students for each mode.

Fig. 4 Frequencies of the additional questions asked by the students



5.2 RQ2. How do different types of questions students ask LLMs influence their learning outcomes?

Using clustering analysis and the elbow method (Hancer & Karaboga, 2017), we identified two student groups based on the patterns of their interaction with ChatGPT: Group 1 (predominantly Passive) and Group 2 (predominantly Active and Constructive). Table 4 summarizes the descriptive statistics of both groups by ICAP question types. As shown, the two clusters were primarily differentiated by large differences in Passive, Active, and Constructive question frequencies, which were the key variables driving the cluster formation, whereas Interactive questions contributed minimally to cluster separation. A MANOVA was then run to compare the differences between the groups on the four types of questions. Given violations of homogeneity of variance-covariance matrices, Pillai's Trace was used as the multivariate test statistic. Results indicated a significant multivariate effect of group, Pillai's Trace = 0.874, $F(4, 35) = 60.84$, $p < .001$. Significant differences between the two groups were found for the frequency of Passive ($F = 63.74$, $p < .001$, $\eta^2 = 0.626$), Active ($F = 66.83$, $p < .001$, $\eta^2 = 0.638$), and Constructive questions ($F = 12.61$, $p = .001$, $\eta^2 = 0.249$). Conversely, no significant difference was found for Interactive questions ($p = .341$). These findings validate the grouping obtained from the clustering analysis, with marked differences in cognitive engagement between the two groups.

A repeated measures ANOVA was then performed to compare the academic performance of the two groups based on Exam 2, administered prior to interaction with ChatGPT, and the Final Exam, which was conducted after this interaction. The assumptions of independence of observations and normality of the dependent variable were met, and since the within-subject factor only included two levels, the sphericity assumption was inherently satisfied (Armstrong, 2017). The findings revealed a statistically significant interaction effect between time and group membership ($F = 4.607$, $p = .038$, $\eta^2 = 0.108$). In particular, Group 2 students improved from Exam 2 (Mean = 17.94) to the Final Exam (Mean = 19.24), while Group 1 students slightly declined from Exam 2 (Mean = 20.13) to the Final Exam (Mean = 18.83). This indicates that students who were subjected to higher-order cognitive questioning (Active and Constructive) benefited more from their interaction with ChatGPT.

Table 4 Rubric for analyzing critical thinking strategies used by students

Strategy	Description	Total
Textbook-Specific	Students cited a specific chapter or section of their textbook as a reference	14 (3.81%)
Textbook-General	Students cited an unnamed chapter or section of their textbook as a reference	13 (3.54%)
Textbook-Class Discussion	Students cited a class discussion on a chapter from their textbook as a reference	22 (5.99%)
Lecture-Specific	Students cited a specific class lecture as a reference	52 (15.53%)
Lecture-Range	Students cited multiple specific class lectures as a reference	3 (0.82%)
Lecture-Non-Specific	Students cited an unnamed lecture or lectures as a reference	5 (1.36%)
Multiple Sources	Students cited multiple types of sources as a reference	18 (4.90%)
Anecdotal Experience	Students cited hearsay, ChatGPT, or personal experience as a reference	18 (4.90%)
No Reference	Students provided no reference	182 (49.59%)

5.3 RQ3. What strategies do students use to evaluate the information received from the LLM?

As it stands, the manner in which the students analyzed the information given to them by ChatGPT can be divided into five main categories, as shown in Table 4. These five were organized according to the sources that the students used to verify ChatGPT's answers and back up their assertion of agreeing or disagreeing: Textbook, Lecture, Multiple Sources, Anecdotal Experience, and No Reference. The names of the categories openly illustrate how the students sought to support their choice to agree or disagree with ChatGPT where Textbook consisted of any time that a student used their textbook as a primary reference, Lecture involves the students using a previous classroom lecture as a primary reference, Multiple Sources consists of when Students cited multiple types of sources as a reference, Anecdotal Experience contains all of the students who cited hearsay, ChatGPT, or personal experience as a reference, and No Reference illustrates the students who provided no reference, but it was necessary to divide both Textbook and Lecture into a set of subcategories to cover the specifics of the students' usage of references. Beginning with Textbook, there are three subcategories: Specific Section, where students cited a specific chapter or section of their textbook as a reference, General Reference, where students cited an unnamed chapter or section of their textbook as a reference, and Class Discussion on Chapter, where students cited a class discussion on a chapter from their textbook as a reference. Likewise, Lecture was also divided into three such subcategories: Specific Lecture Number, where students cited a specific class lecture as a reference; Range of Lec-

tures, where students cited multiple specific class lectures as a reference; and Non-Specific Lecture, where students cited an unnamed lecture or lectures as a reference. Additionally, these subcategories and all major categories have been further divided according to whether the student agreed or disagreed with the response of ChatGPT.

The two outliers, “Multiple Sources” and “No Reference,” occupy the low and high ends of the distribution, respectively. Although Multiple Sources represent a minority (18 or 4.9%), Textbook (49 or 13.35%), Lecture (60 or 16.35%), and Anecdotal Experience (58 or 15.80%) consist of a population relative to each other. However, No Reference, which is by far the largest group (182 or 49.59%), pulls away from every other category with a large lead. Furthermore, students agreed with ChatGPT more often than not in all categories of evaluation in 324 agreement responses (88.28%), while students only disagreed with 43 of the ChatGPT responses (11.72%).

6 Discussions and implications

Our findings highlight challenges in students’ engagement with AI-driven learning, particularly their limited interaction with LLMs, their low levels of cognitive engagement in conversations, and their reliance on anecdotal experiences rather than evidence-based reasoning when it comes to assessing the reliability of LLM-generated responses. These patterns have important implications for AI literacy and instructional strategies aimed at fostering deeper engagement with GenAI.

6.1 High proportion of students who do not ask additional questions

One of the most striking findings was that a significant proportion of students did not ask additional questions beyond the initial prompt provided as instruction (48.6%), indicating a passive approach to LLM-assisted learning. This high proportion of students who did not ask additional questions suggests that many participants may not yet understand how to leverage ChatGPT for deeper inquiry. A study by Grande et al. (2024) found that when students used an LLM of their choice as a partner for an in-class ethics discussion, they tended to view LLMs as an “authoritative knowledge base” as opposed to a tool that could be questioned and used to practice argumentation and critical thinking. This aligns with previous research, which has shown that students often use LLMs as a source of quick answers rather than as a tool for exploration and reasoning (Al-Maroofo et al., 2024).

Since most of the students in our study had never used ChatGPT before, their lack of engagement in iterative questioning can likely be attributed to several factors related to this unfamiliarity. One key factor is that students may not fully understand how LLMs function as learning tools, leading them to use ChatGPT in the same way they would use a search engine, which simply retrieves information rather than engages in a dialogue that fosters deeper learning. This aligns with prior findings that students often perceive ChatGPT as equivalent to Google (Ding et al., 2023), failing to recognize its potential for interactive reasoning and knowledge construction. Additionally, students may assume that LLM-generated responses are inherently accurate and complete, which may explain why a large proportion of them

accepted ChatGPT's responses without further verification. This assumption of reliability is evident in our data, where most students did not provide references when agreeing with ChatGPT's responses, suggesting that they placed unwarranted trust in the AI's outputs. LLM tools such as ChatGPT are increasingly perceived as "epistemic authorities" whose accuracy goes unchecked owing to their authoritative tone and human-like interactions (Urban et al., 2025). Without prior exposure to LLMs, students may not be aware of their limitations, such as hallucinations and biases, and as a result, may not see the need to critically evaluate the information provided. This finding also sheds light on the fact that students have varying levels of exposure, access, and knowledge of LLMs, which in turn shapes their disposition and criticality towards these tools.

While LLMs can generate rapid responses, prior research emphasizes that they cannot substitute for students' critical thinking and problem-solving abilities (Dwivedi et al., 2023). Our findings underscore the need for the "Initiate" and "Interact" phases of the HIIC model, where students are expected to initiate and continue the dialogue with the LLM, rather than stopping after receiving an initial response. Our findings suggest that AI literacy initiatives in similar instructional contexts could extend beyond basic PE and explicitly educate students on the affordances and limitations of LLMs (e.g., language coherence vs. hallucination effects), as well as how to engage in dialogue with LLMs to refine and deepen their understanding. This also includes teaching students to engage critically with LLMs (i.e., the "Critique" in the HIIC model) and uphold intellectual integrity as a part of their AI literacy skills (Idris et al., 2024). Educators could integrate structured LLM-assisted inquiry activities at the beginning of a semester to teach students how to ask follow-up questions, evaluate multiple LLM responses, and compare LLM-generated explanations with course materials.

6.2 Low levels of cognitive engagement in AI interactions

Even among students who engaged with ChatGPT, most interactions remained at the active level, with only a small fraction reaching constructive or interactive engagement. Students primarily asked ChatGPT to define terms or explain why incorrect answers were wrong rather than formulating deeper, exploratory questions that would push their understanding forward. The low proportion of interactive-mode questions (4.1%) suggests that students are not yet using LLMs as a tool for collaborative reasoning. Moreover, interactions that are limited to passive and active levels may foster dependence on LLM technology and set off "metacognitive laziness," wherein students offload cognitive tasks critical to learning onto LLMs, ultimately hampering their ability to self-regulate and engage deeply in the learning process (Fan et al., 2025).

From an ICAP perspective, this pattern likely reflects insufficient external scaffolding to prompt generative or dialogic processing, leading students to default to lower-effort, information-seeking behaviors rather than knowledge construction (Chi & Wylie, 2014). Future research can borrow the techniques of scaffolding question-asking in traditional classrooms to train students how to engage in more constructive and interactive interactions with LLMs. For example, one way to encourage

constructive and interactive engagement is to train students in question-refinement techniques, helping them develop progressively deeper questions that build on AI responses. Educators could model progressive questioning strategies, which support the “Interact” phase of the HIIC model by encouraging sustained, generative engagement with LLMs. When teaching how to interact with LLMs, the instructors could encourage students to ask more than one question on the same topic, ask at least two follow-up questions that contain their own reasoning, critiques, or understanding before accepting an AI’s response, and scaffold students to move from basic factual questions to “what if” and “how might” questions via question templates or provide examples of meaningful interactive questions, etc. Essel et al. (2024) found that students showed significant improvement in critical, creative, and reflective thinking when using ChatGPT, especially when supported with scaffolds in the form of ‘didactic assistance’ during in-class activities. Techniques such as role playing, debating, quizzing, etc., could also be taught to students to help them facilitate meaningful conversations with LLMs to benefit their learning. By explicitly teaching these strategies, students can move from simply extracting information to engaging in a process of co-construction with LLMs, aligning more closely with constructive and interactive learning modes of ICAP.

6.3 Overreliance on anecdotal and unverified sources

Another important finding was that nearly half of the students (49.6%) did not reference any external sources when evaluating the responses of ChatGPT. Furthermore, a significant number of students relied on anecdotal evidence or personal intuition rather than authoritative sources such as textbooks or lecture materials. This suggests that many students may struggle to evaluate the credibility of AI-generated information, a critical skill in an era of AI-generated misinformation. Idris et al. (2024) warn that overreliance on LLM-generated content without engaging in validation and critical evaluation may erode students’ critical thinking skills.

This pattern is particularly concerning given that GenAI is prone to hallucinations (Farquhar et al., 2024), which means that students who do not critically assess the LLM responses risk internalizing incorrect information. For students to fully leverage the power of LLMs in supporting their learning, they need to adopt the humans are essential mindset and be equipped with foundational knowledge of AI literacy and critical thinking skills, such as how to cross-reference LLM outputs with textbooks, peer-reviewed sources, and credible databases (Shneiderman, 2022). This aligns with the “Critique” phase of the HIIC model, which underscores the human responsibility to verify and judge AI-generated information rather than accept it at face value. One practical instructional strategy could involve structured LLMs fact-checking exercises, where students are required to compare LLMs’ responses to at least two credible sources before making a judgment. Additionally, embedding case studies of AI misinformation into curricula could raise awareness of AI limitations and foster a more critical, evidence-based approach to AI-assisted learning.

6.4 The influence of cognitive engagement on academic performance

Contrary to previous studies, which raised concerns that the use of LLMs can negatively influence students' learning due to over-reliance (Bastani et al., 2024), we found that students' learning outcomes were positively associated with how they engaged with ChatGPT. The key difference between our findings and previous studies is that we found that students who posed deeper questions outperformed students who posed more superficial questions, but previous studies compared overall use of LLMs without considering initial engagement levels. Students who primarily posed active and constructive questions (Group 2) showed an improvement between Exam 2 and the Final Exam. In comparison, students who primarily engaged in passive questioning failed to exhibit improved performance, reflecting the importance of promoting deeper cognitive engagement in integrating LLMs in classrooms. Interestingly, students in the predominantly passive group scored higher on Exam 2 but declined by the Final Exam, whereas students who engaged more actively and constructively showed greater improvement over time. This pattern may reflect differences in perceived need for engagement, with higher-performing students relying more on passive confirmation and lower-performing students engaging more actively to repair misunderstandings; however, these findings are correlational rather than causal.

This discovery has significant implications for the development of teaching frameworks. Based on the patterns observed in this study, instructors may consider creating targeted interventions that prompt students to interact more actively and constructively with LLMs, for instance, through guided prompts that necessitate deep thinking or by utilizing iterative questioning strategies. A study by Wu et al. (2024) found that current AI instructional practices inadequately foster AI literacy, with only 35.71% addressing higher-order skills such as evaluating and creating with AI. Subsequent research must look into how best to initiate the shift of students from passive to more cognitively challenging interactions with LLMs. Given the inevitability of future human-LLM interactions and partnerships in education, Chen and Zhu (2023) underscore the importance of supporting and cultivating mindful engagement characterized by reflection, effort, and metacognition. Furthermore, these findings also suggest that educators, such as preservice and in-service teachers, need explicit preparation on how to model and scaffold productive student-LLM interactions, rather than assuming students will naturally engage in inquiry-based questioning when using AI tools.

7 Conclusions

In this study, we used the HIIC model to guide the analysis of students' interactions with LLMs. The results highlight two critical challenges in students' interactions with LLMs: low engagement in constructive and interactive conversations and a lack of critical evaluation of LLM-generated responses. Nearly half of the students did not ask additional questions, and those who did primarily engaged at an active rather than constructive or interactive level. Additionally, most students failed to verify LLM

responses with credible sources, relying instead on anecdotal experiences or personal intuition. The cluster analysis emphasized the significance of cognitive engagement patterns, which reflected apparent differences in learning outcomes depending on the quality of interactions students had with ChatGPT. Students who engaged actively with the content by creating constructive questions had their exam scores improve, which reflects the significance of stimulating higher-order cognitive engagement with AI tools.

To maximize the educational potential of LLMs, students must be explicitly taught how to engage in deeper, iterative conversations that build knowledge rather than passively retrieve information. Instructional strategies should intentionally stimulate and support active and constructive interactions. Equally important is training students to critically evaluate LLM-generated content, ensuring they develop the reasoning skills needed to assess accuracy and reliability. For learning analytics research, this study demonstrates how theory-driven frameworks such as ICAP can be used to analyze fine-grained student-LLM interaction data and distinguish meaningful engagement patterns linked to learning outcomes. The HIIC model offers a conceptual foundation for designing instructional strategies that move beyond prompt engineering and support deeper, more intentional engagement with LLMs. Future research should focus on instructional interventions that scaffold these skills, equipping students with the tools to collaborate effectively with LLMs while maintaining a critical mindset.

8 Limitations and future directions

While this study provides valuable insights into students' engagement patterns with LLMs, it has several limitations that should be considered when interpreting the results. The study relied on students' submitted responses from the makeup exam assignment, which may not fully capture the entirety of their interactions with LLMs. Some students may have engaged in additional questioning outside the recorded responses, but these interactions were not systematically collected. Future studies should consider directly logging LLM interaction data to provide a more comprehensive view of students' engagement behaviors. Secondly, although the ICAP framework provides a useful lens for characterizing observable engagement behaviors, its application in LLM-mediated contexts has inherent limitations. In particular, students' interactions with LLMs are partially mediated by AI-generated responses, which may mask or amplify underlying cognitive processes and make it difficult to fully infer learners' internal reasoning from surface-level question forms alone. Future research should explore complementary methods, such as think-aloud protocols or process-tracing approaches, to more directly capture students' cognitive engagement when interacting with LLMs. Additionally, we did not examine the accuracy or quality of ChatGPT's responses; future research could directly analyze how incorrect, repetitive, or unhelpful LLMs' outputs influence students' learning and critical engagement.

Third, this study analyzed student interactions descriptively but did not directly measure learning outcomes resulting from different engagement modes. In addition, other factors that may influence exam performance, such as students' motivation or

study time, were not measured in this study. Future research should employ controlled experiments that compare instructional interventions aimed at increasing constructive and interactive engagement with LLMs and assess their impact on student learning. Additionally, the HIIC model presented in this study is conceptual in nature and should be treated as a preliminary framework. Future work should aim to conduct controlled experimental studies to empirically examine its effectiveness in guiding student-AI interactions for deeper learning.

Fourth, almost all students who participated in this study had little to no knowledge of AI literacy, which may significantly influence how they evaluate LLM-generated responses. Students with more experience using AI tools may demonstrate greater critical thinking, while those with limited exposure may be more likely to accept the responses at face value; future research should examine how prior AI knowledge shapes engagement patterns and verification behaviors. Last but not least, caution needs to be exercised when applying the results directly to another physics course. The sample in this study was relatively small and predominantly female, which contrasts with typical gender distributions in physics education, where female enrollment tends to be much lower. This imbalance reflects the context of the course, which was an introductory physics class designed for non-majors and may attract a different demographic than traditional STEM pathways. Future research should examine this with a larger sample size and should examine whether gender or other demographic variables influence the way students engage with LLMs for learning, particularly in domains where such disparities are common.

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Declarations

Competing Interests The authors declare that they have no competing interests.

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